BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Objective Type (Group-I)

Time Allowed: 30 Min. | **Maximum Marks:** 20

Note: Inter, Part I, Lahore, 2021. Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. The number of roots of polynomial equation $8x^6 - 19x^3 - 27 = 0$ are:

(A) 2 (B) 4

(C) 6 (D) 8

Q2. For an infinite geometric series of which $|r| < 1$, we have S_∞ =:

(A) $a(1+r)/(1-r)$ (B) $a/(1+r)$

(C) $a/(2r)$ (D) $a/(1-r)$

Q3. The property used in $(a+1)+3/4 = a+(1+3/4)$ is:

(A) Closure (B) Associative

(C) Commutative (D) Additive

Q4. If $a_{n-3} = 2n - 5$, then 7th term is =:

(A) 9 (B) 15

(C) 11 (D) 13

Q5. The converse of $p \rightarrow q$ is:

(A) $\sim p \rightarrow q$ (B) $p \rightarrow \sim q$

(C) $q \rightarrow p$ (D) $\sim p \rightarrow \sim q$

Q6. Partial fraction of $1/(x^2-1)$ will be in of the form:

(A) $(Ax+B)/(x^2-1)$ (B) $A/(x+1) + B/(x-1)$

(C) $A/(x+1)$ (D) $B/(x-1)$

Q7. When $p(x) = x^3 + 4x^2 - 2x + 5$ is divided by $(x - 1)$, then remainder is:

(A) 10 (B) -10

(C) 8 (D) -8

Q8. If $|k\ 4;\ 4\ k| = 0$, then value of k is:

(A) ± 16 (B) 0

(C) ± 4 (D) ± 8

Q9. If H is H.M. between a and b, then H =:

(A) $2ab/(a+b)$ (B) $(a+b)/(2ab)$

(C) $(a+b)/2$ (D) $\pm\sqrt{ab}$

Q10. The trivial solution of the homogeneous linear equation in three variables is:

(A) (0,0,0) (B) (1,0,0)

(C) (0,1,0) (D) (0,0,1)

Q11. The period of $3\cos(x/5)$ is:

(A) π (B) 10π

(C) $\pi/10$ (D) $\pi/5$

Q12. The factorial of a positive integer 'n' is:

(A) $n! = n(n-1)!(n-2)!$ (B) $n! = n(n+2)!$

(C) $n! = n(n-1)!$ (D) $n! = n(n-2)!$

Q13. The solution of $1 + \cos x = 0$ if $0 < x \leq 2\pi$ is equal to:

(A) $\{0\}$ (B) $\{\pi/2\}$

(C) $\{\pi/3\}$ (D) $\{\pi\}$

Q14. $\cos 48° + \cos 12° =$:

(A) $2\cos 30° \cos 18°$ (B) $3\cos 18°$

(C) $\sqrt{3}\cos 18°$ (D) $\sqrt{2}\cos 18°$

Q15. $\sec[\cos^{-1}(1/2)] =$:

(A) 1/2 (B) 2

(C) $\pi/3$ (D) $\pi/6$

Q16. The middle term in expansion of $(a+x)^n$ when n is even:

(A) $(n/2+1)^{th}$ term (B) $(n/2-1)^{th}$ term

(C) $(n/2)^{th}$ term (D) $((n+1)/2)^{th}$ term

Q17. $9\pi/5$ rad in degree measure is:

(A) 321° (B) 322°

(C) 323° (D) 324°

Q18. If Δ is the area of a triangle ABC, then Δ =:

(A) ½ bc sinβ (B) ½ ab sinα

(C) ½ bc sinα (D) ab sinα

Q19. In anti-clockwise direction, ¼ rotation is equal to:

(A) 90° (B) 180°

(C) 270° (D) 45°

Q20. With usual notations, the value of a + b + c is =:

(A) s (B) 2s

(C) 3s (D) s/2

Essay Type (Group-I)

Time Allowed: 2.30 hrs. | **Maximum Marks:** 80

QSECTION - I. (Marks 16)

QQ.2. (Marks 16) Write short answers to any EIGHT (8) questions:

i Prove that a/b = ka/kb, k ≠ 0

ii Simplify (5,−4) ÷ (−3,−8) and write the answer as a complex number.

iii Find the real and imaginary parts of $(\sqrt{3} + i)^3$

iv If B = {1,2,3}, then find the power set of B, i.e., P(B).

v Construct the truth table for the statement: $\sim(p \rightarrow q) \leftrightarrow (p \wedge \sim q)$

vi For the set A = {1,2,3,4}, find a relation in A which satisfy {(x,y)| y + x = 5}

vii Find the matrix X, if 2X + 3A = B and A = [2 1 2; −2 4 5], B = [3 −1 0; 4 2 1]

viii Find A^{-1} if A = [5 3; 1 1]

ix Without expansion, show that |α β+γ 1; β γ+α 1; γ α+β 1| = 0

x Prove that sum of cube roots of unity is zero i.e., $1 + \omega + \omega^2 = 0$

xi Find the numerical value of k, when the polynomial $x^3 + kx^2 - 7x + 6$ has a remainder of -4 when divided by $x + 2$.

xii Show that the roots of equation $x^2 + (mx + c)^2 = a^2$ will be equal if $c^2 = a^2(1 + m^2)$.

QQ.3. (Marks 16) Write short answers to any EIGHT (8) questions:

i Resolve $4x^2/((x^2+1)^2(x-1))$ into partial fractions.

ii Resolve $(7x+25)/((x+3)(x+4))$ into partial fractions.

iii Write the first four terms of the sequence, $a_n = (-1)^n n^2$.

iv If $a_{n-3} = 2n - 5$, find nth terms of the sequence.

v Insert two G.M's between 2 and 16.

vi Sum the infinite geometric series 1/2 + 1/4 + 1/8 + ...

vii Find the value of n, when ${}^nP_n = 1.10.9$.

viii Evaluate ${}^{12}C_3$.

ix A die is rolled. What is the probability that the dots on the top are greater than 4?

x Check the truth of the statement $1 + 5 + 9 + ... + (4n - 3) = n(2n - 1)$ for n = 1, 2.

xi Calculate by means of binomial theorem $(2.02)^4$.

xii If x is so small that its square and higher powers can be neglected, then show that $\sqrt{(1+2x)}/\sqrt{(1-x)} \approx 1 + 3/2\ x$.

QQ.4. (Marks 18) Write short answers to any NINE (9) questions:

i Convert 54° 45′ into radians.

ii If $\cot\theta = 15/8$ and the terminal arm of the angle is not in quadrant I, find the value of cosec θ.

iii Verify $2 \sin 45° + ½ \operatorname{cosec} 45° = 3/\sqrt{2}$.

iv Prove that $\cot(\alpha + \beta) = (\cot\alpha \cot\beta - 1)/(\cot\alpha + \cot\beta)$

v Prove that $\tan(180° + \theta) = \tan\theta$

vi Express $2 \sin 7\theta \sin 2\theta$ as sums or differences.

vii Find the period of tan x/7.

viii A vertical pole is 8 m high and the length of its shadow is 6 m. What is the angle of elevation of the Sun at that moment?

ix Find area of the triangle ABC if a = 200, b = 120, $\gamma = 150°$.

x Prove that $r\ r_1\ r_2\ r_3 = \Delta^2$.

xi Find the value of $\sec(\sin^{-1}(-1/2))$.

xii Show that $r = (s - a)\tan(\alpha/2)$.

xiii Find the solution of $\operatorname{cosec}\theta = 2$ which lies in the interval $[0, 2\pi]$.

QSECTION - II. Note: Attempt any THREE (3) questions.

Q5a. (Marks 5) Solve by Cramer's rule $2x_1 - x_2 + x_3 = 8$, $x_1 + 2x_2 + 2x_3 = 6$, $x_1 - 2x_2 - x_3 = 1$

Q5b. (Marks 5) If α, β are roots of equation $ax^2 + bx + c = 0$, form the equation whose roots are $\alpha + 1/\alpha$, $\beta + 1/\beta$.

Q6a. (Marks 5) Resolve $(3x - 11)/((x^2 + 1)(x + 3))$ into partial fraction.

Q6b. (Marks 5) If $S_n = n(2n - 1)$, then find the series.

Q7a. (Marks 5) Prove that ${}^{n-1}C_r + {}^{n-1}C_{r-1} = {}^{n}C_r$.

Q7b. (Marks 5) Use mathematical induction to prove $(3C3) + (4C3) + (5C3) + \ldots + ({}^{n+2}C3) = ({}^{n+3}C4)$ for every positive integers n.

Q8a. (Marks 5) Two cities A and B lie on the equator, such that their longitudes are 45° E and 25° W, respectively. Find the distance between the two cities, taking the radius of the earth as 6,400 kms.

Q8b. (Marks 5) Prove that $(\sin\theta + \sin 3\theta + \sin 5\theta + \sin 7\theta)/(\cos\theta + \cos 3\theta + \cos 5\theta + \cos 7\theta) = \tan 4\theta$.

Q9a. (Marks 5) Solve the triangle ABC, if $a = 53$, $\beta = 88° 36'$, $\gamma = 31° 54'$.

Q9b. (Marks 5) Prove that $\tan^{-1} 1/11 + \tan^{-1} 5/6 = \tan^{-1} 1/3 + \tan^{-1} 1/2$.

GROUP – II

Objective Type (Group-II)

Time Allowed: 30 Min. | **Maximum Marks:** 20

Note: Inter, Part I, Lahore, 2021. Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. If x is a cube root of unity then $x^2 + x + 1 =$:

(A) 0 (B) 1

(C) 2 (D) 3

Q2. $(1 - w^{17})^2 =$:

(A) -4 (B) 9

(C) $-4w$ (D) $-3w^2$

Q3. $(6, -5)^2 =$:

(A) $11 - 60i$ (B) $61 - 60i$

(C) $61 + 60i$ (D) $11 + 60i$

Q4. A is the power set of {a, b}, then n[A] =:

(A) 2 (B) 8

(C) 4 (D) 6

Q5. The converse of the conditional $p \rightarrow q$ is:

(A) $p \rightarrow \sim q$ (B) $\sim p \rightarrow q$

(C) $q \rightarrow p$ (D) $\sim q \rightarrow p$

Q6. The partial fractions of $1/(x^2 + 2x + 1)$ are:

(A) $A/(x+1) + B/(x+1)^2$ (B) $A/(x+1)$

(C) $(Ax+B)/(x^2+2x+1)$ (D) $A/(x+1)^2 + B/(x+1)^2$

Q7. If $a, b \in R$ then $a + b \in R$ is the property of:

(A) Commutative (B) Closure

(C) Associative (D) Additive inverse

Q8. If A is a square matrix and $A^t = A$ then A is:

(A) Symmetric matrix (B) Skew-symmetric matrix

(C) Diagonal matrix (D) Singular matrix

Q9. If $3C_3 + 3C_4 = 4C_r$, then r =:

(A) 2 (B) 1

(C) 0 (D) 5

Q10. [illegible in source scan]

(A) $8C_2$ (B) $8P_2$

(C) $8P_8$ (D) $8C_8$

Q11. The period of $3\cos(2x)$ is:

(A) 2π (B) 3π

(C) π (D) 6π

Q12. The solution set of $1 - \cos x = 0$ if $x \in [0, 2\pi]$ is:

(A) $\{0\}$ (B) $\{2\pi\}$

(C) $\{0, 2\pi\}$ (D) $\{\pi\}$

Q13. $\tan(360° - \theta) =$:

(A) $\tan \theta$ (B) $-\tan \theta$

(C) $\cot \theta$ (D) $-\cot \theta$

Q14. $3\pi/5$ rad in degree measure is:

(A) 108° (B) 109°

(C) 110° (D) 111°

Q15. The general solution of $\cot x = 0$ is:

(A) $x = n\pi$ (B) $x = (2n + 1)\pi/2$

(C) $x = (n + 1/2)\pi$ (D) $x = n\pi/2$

Q16. In a triangle ABC, the value of s =:

(A) a+b−c/2 (B) a+b+c/2

(C) a−b+c/2 (D) −a+b+c/2

Q17. In a triangle ABC, if a = 3, b = 4, c = 5, then r =:

(A) 1 (B) 2

(C) 3 (D) 4

Q18. The solution set of equation $\sin^2 x - \sin x = 0$ is:

(A) $\{x: x = n\pi, n \in Z\}$ (B) $\{x: x = n\pi, n \in Z\} \cup \{x: x = 2n\pi + \pi/2, n \in Z\}$

(C) $\{x: x = n\pi, n \in Z\} \cup \{x: x = 2n\pi - \pi/2, n \in Z\}$ (D) $\{x: x = n\pi, n \in Z\} \cup \{x: x = 2n\pi \pm \pi/2, n \in Z\}$

Q19. In triangle ABC, the value of $\sin(\alpha/2)$ is:

(A) √[(s−b)(s−c)/bc] (B) √[(s−b)(s−a)/ac]

(C) √[(s−c)(s−a)/ab] (D) √[(s−a)(s−b)/ac]

Q20. If 2 is the arithmetic mean of √2 and √8, then the common difference is:

(A) √2 (B) 0

(C) −√2 (D) 4

Essay Type (Group-II)

Time Allowed: 2.30 hrs. | **Maximum Marks:** 80

QSECTION - I. (Marks 16)

QQ.2. (Marks 16) Write short answers to any EIGHT (8) questions:

i Separate into real and imaginary parts $(2-7i)/(4+5i)$.

ii Prove that for $\forall z \in c$ $z.\bar{z}=|z^2|$.

iii Find out real and imaginary parts of complex number $(\sqrt{3}+i)^3$.

iv If G be a group and a, b ∈ G, then show that $(ab)^{-1}=b^{-1}a^{-1}$.

v Give a table for addition of elements of the set of residue classes modulo 5.

vi Show that $(p \wedge q) \rightarrow p$ is a tautology.

vii Find x and y if $[x+3\ 1; -2\ 3y-4] = [y\ 1; -3\ 2x]$.

viii Find the inverse of $[-2\ 3; -4\ 5]$.

ix Without expansion verify that $|bc\ ca\ ab; 1/a\ 1/b\ 1/c; a\ b\ c| = 0$.

x Convert $x^{12} - x^{14} - 6 = 0$ into quadratic equation.

xi Evaluate $(-1+\sqrt{-3})^5 + (-1-\sqrt{-3})^5$.

xii Discuss the nature of the roots of $2x^2 - 5x + 1 = 0$

QQ.3. (Marks 16) Write short answers to any EIGHT (8) questions:

i Write $1/((1-ax)(1-bx)(1-cx))$ into partial fractions.

ii Write $4x^2/((x^2+1)^2(x-1))$ into partial fractions.

iii If $a_{n-3} = 2n - 5$, find nth term of the sequence.

iv Find G.M. between -2i and 8i.

v If the numbers $1/k$, $1/(2k+1)$, $1/(4k-1)$ are in H.P., find the value of k.

vi Find A, G and H if a = 2i, b = 4i.

vii Find the value of n when ${}^nP_2 = 30$.

viii Find the number of the diagonals of a 6-sided figure.

ix A die is rolled. What is the probability that the dots on the top are greater than 4?

x Calculate $(9.98)^4$ by using binomial theorem.

xi Expand $(4-3x)^{1/2}$ up to 4 terms by using binomial theorem.

xii Evaluate ${}^{12}C_3$.

QQ.4. (Marks 18) Write short answers to any NINE (9) questions:

i Convert 75°6′30″ into radians.

ii Evaluate $(1-\tan^2(\pi/3))/(1+\tan^2(\pi/3))$

iii Prove that $\sec^2 A + \operatorname{cosec}^2(A) = \sec^2(A)\operatorname{cosec}^2(A)$ where $(A \neq n\pi/2, n \in z)$.

iv Prove that $\tan(180° + \theta) = \tan\theta$.

v Prove that $\cot(\alpha+\beta) = (\cot\alpha \cot\beta - 1)/(\cot\alpha + \cot\beta)$

vi Prove that $\sin 2\alpha/(1+\cos 2\alpha) = \tan\alpha$.

vii Find the period of $\tan x/7$.

viii In ΔABC if $\beta = 60°$, $\gamma = 15°$ and $b = \sqrt{6}$, then find 'c'.

ix In ΔABC if $a = 34$, $b = 20$ and $c = 42$, find angle 'r'.

x Show that $r = (s-a)\tan(\alpha/2)$.

xi Show that $\cos^{-1}(-x) = \pi - \cos^{-1}(x)$

xii Find the value of $\sec(\sin^{-1}(-1/2))$

xiii Find the solution of $\operatorname{cosec}\theta = 2$ which lie in $[0, 2\pi]$

QSECTION-II. Note: Attempt any THREE (3) questions.

Q5a. (Marks 5) Solve the system of equations by Cramer's rule $2x+2y+z=3$, $3x-2y-2z=1$, $5x+2y-3z=2$.

Q5b. (Marks 5) Solve the system of equations. $2x - y = 4$; $2x^2 - 4xy - y^2 = 6$

Q6a. (Marks 5) Resolve $(x-1)/((x-2)(x+1)^3)$ into partial fraction.

Q6b. (Marks 5) Find four A.Ms between $\sqrt{2}$ and $12/\sqrt{2}$.

Q7a. (Marks 5) Find the values of n and r when ${}^nC_r = 35$ and ${}^nP_r = 210$.

Q7b. (Marks 5) Find the term involving x^4 in the expansion of $(3-2x)^7$.

Q8a. (Marks 5) Prove that $(1+\cos\theta)/(1-\cos\theta) = (\operatorname{cosec}\theta + \cot\theta)^2$

Q8b. (Marks 5) Prove that $\cos 3\theta/\cos\theta + \sin 3\theta/\sin\theta = 4\cos 2\theta$.

Q9a. (Marks 5) Prove that $(r_1 + r_2)\tan(\gamma/2) = c$.

Q9b. (Marks 5) Prove that $\sin^{-1} 5/13 + \sin^{-1} 7/25 = \cos^{-1} 253/325$.